

# Making Computer Graphics History Public

## SIGGRAPH 2021 Retrospective Panel

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## 1 INTRODUCTION

This panel concentrates on publications in diverse formats, mainly books and video documentaries, that make public the research about the history of computer graphics and interactive techniques. Both of the books presented in the panel have been published recently by MIT Press, making this a unique opportunity to listen to the authors: The panel has three presentations by key scholars and researchers:

### 1.1 “Image Objects” — Book presentation by Jacob Gaboury

This talk will introduce my new book on the early history of computer graphics titled *Image Objects: An Archaeology of Computer Graphics* (MIT Press, 2021). The book explores the important role that computer graphics played in the development of computer science in the second half of the twentieth century, with a focus on the archives and historical collections of the research program at the University of Utah from roughly 1965-1980. Adopting an “object-oriented” approach, I explore five technologies produced by Utah faculty and alumni that fundamentally reshaped the field

of computer science and helped to transform the computer from a calculating machine into an interactive medium.

### 1.2 “A Biography of the Pixel” — Book presentation by Alvy Ray Smith

The Great Digital Convergence of all media types into one universal digital medium occurred, with little fanfare, at the recent turn of the millennium. The bit became the universal medium, and the pixel—a particular packaging of bits—conquered the world. Henceforward, nearly every picture in the world would be composed of pixels—cell phone pictures, app interfaces, Mars Rover transmissions, book illustrations, videogames. In *A Biography of the Pixel*, Pixar cofounder Alvy Ray Smith argues that the pixel is the organizing principle of most modern media, and he presents a few simple but profound ideas that unify the dazzling varieties of digital image making. Alvy’s story of the pixel’s development begins with Fourier waves, proceeds through Turing machines, and ends with the first digital movies from Pixar, DreamWorks, and Blue Sky. Today, almost all the pictures we encounter are digital—mediated by the pixel and irretrievably separated from their media; museums and kindergartens are two of the last outposts of the analog. Alvy explains, engagingly and accessibly, how pictures composed of invisible stuff become visible—that is, how digital pixels convert to analog display elements. Taking the special case of digital movies to represent all of Digital Light, and drawing on his decades of work in the field, Alvy approaches his subject from multiple angles—art, technology, entertainment, business, and history. *A Biography of the Pixel* is aimed at anyone who has watched a video on a cell phone, played a videogame, or seen a movie.

### 1.3 “SIGGRAPH @50 History” by Mary Whitton, Adele Newton, and Dave Kasik

As part of the advanced work for the ACM SIGGRAPH Conference’s 50th anniversary in 2023, the SIGGRAPH History Committee is conducting interviews a wide range of graphics pioneers. Their personal stories trace how they got involved in computer graphics and contributed to the Conference’s storied past. Their stories provide insight into drivers for long careers and capture their passion. The contributors share stories about graphics, graphics devices, and

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applications that drove innovation in engineering, science, industrial design, education, and any field that uses images as part of its vocabulary. The oral histories complement the other panelists'; excellent books.

## 2 BIOGRAPHIES

*Jacob Gaboury.* Jacob Gaboury works in the areas of digital media, visual culture, media archaeology, and queer theory. His teaching and research interests include 20th century histories of technology and computation, queer and feminist science and technology studies, computer graphics and digital game studies, and the intersection of contemporary art and technology. His forthcoming book, titled *Image Objects* (MIT Press), offers a material history of early computer graphics in the United States told through a set of five technical objects that shape and structure the production of most all contemporary digital images. His work has appeared in a wide range of popular and academic publications, including the *Journal of Visual Culture*, *Grey Room*, *Camera Obscura*, *Women & Performance*, *Debates in the Digital Humanities*, *Rhizome*, and *Art Papers*.

*Dr. Alvy Ray Smith.* Dr. Alvy Ray Smith cofounded two successful startups: Pixar - see Pixar founding documents - (sold to Disney) and Altamira (sold to Microsoft). First director of computer graphics at Lucasfilm. Original member of the Computer Graphics Lab of the New York Institute of Technology. First Graphics Fellow at Microsoft. At Xerox PARC for the birth of the personal computer, the internet, and some of the earliest color pixels. Received two technical Academy Awards, for the alpha channel and digital paint systems. Invented the first full-color paint program, the HSV (or HSB) color transform, and the alpha channel. Directed the Genesis Demo in *Star Trek II: The Wrath of Khan*. Hired John Lasseter and directed him in *The Adventures of André & Wally B.* Proposed and negotiated the Academy-Award winning Disney computer animation production system, CAPS. Instrumental, as a Regent, in initiating the Visible Human Project of the National Library of Medicine. Star witness in a trial that successfully invalidated five patents that threatened Adobe Photoshop. Active in the development of the HDTV standard, arguing for progressive scan. Holds Ph.D. from Stanford University and honorary doctorate from New Mexico State University. Member of the National Academy of Engineering, Fellow of the American Association for the Advancement of Science, and Fellow of the American Society of Genealogists. Founding member of the SIGGRAPH Academy. Published widely in theoretical computer science, computer graphics, and scholarly genealogy. Creator of many pieces of computer art, including *Sunstone* in the collection of New York's Museum of Modern Art. Holds four patents. About to publish a book, *A Biography of the Pixel*, with MIT Press (3 Aug 2021). An advisor to Baobab Studios, an award-winning VR startup in Silicon Valley. For more details see <http://alvyray.com>.

*Mary Whitton.* Mary Whitton attended her first SIGGRAPH Conference in 1977 (San Jose). She was a founder of Ikonas Graphics Systems (1978) and Trancept Systems (1986). Both companies designed and sold high-end user-programmable graphics hardware—think of it as a GPU the size of a dorm refrigerator—and libraries for

graphics, image processing, volume rendering, and visualization. The companies' products were widely adopted in academic and industrial research laboratories for applications including seismic exploration, 3D medical imaging, intelligence analysis, computer animation, and scientific modeling and simulation. After joining UNC in 1994, Whitton co-led the Effective Virtual Environments research group that developed and evaluated technologies to make virtual environment systems more effective for applications such as simulation, training, and rehabilitation. A "mostly-retired" research professor, Whitton spends much of her time these days working on preserving computer graphics history.

*Adele Newton.* Adele Newton developed and managed a broad range of programs and strategic international partnerships in the university, technology and not-for-profit sectors across Canada, within North America and in Africa, Europe and China. Her career has focused on connecting diverse partners from technology, science and the arts to work together on exciting and innovative projects. Her current projects include her work as Executive Director of CS-Can|Info-Can and University Relationship Manager for Profound Impact Corporation. Adele is a graduate of the Faculty of Mathematics at the University of Waterloo. Adele has served in a range of ACM SIGGRAPH roles, including Registration Chair in 1984, Conference Co-Chair for ACM SIGGRAPH 88, member of the Executive Committee as Chair of the Conference Planning Committee and chair of the ACM SIGGRAPH 2019 Business Symposium. She is currently Chair of the ACM SIGGRAPH Communications Committee and member of the SIGGRAPH 50th Conference Anniversary Executive Committee.

*Dave Kasik.* Dave Kasik retired as Boeing's Senior Technical Fellow in visualization and interactive techniques in 2016. In this role, he advanced Boeing's use of digital twin 3D design and digital thread processes. Dave devoted his first 11 years at Boeing to research and development of computer-aided design software and led to pioneering work in interactive 3D graphics, user interface management systems, and industrial use of non-uniform rational b-spline (NURBS) solids and surfaces. Later projects introduced Boeing to Massive Model Visualization and visual analytics. Dave participates in ACM (Association for Computing Machinery), which named him a Fellow in 2013; ACM SIGGRAPH, where he started the annual Pioneers student mentoring program, ran the Exhibits in 1980 and 1994, and received the Outstanding Service Award in 2012; and IEEE, where he is member of the editorial board of *Computer Graphics and Applications*. Dave is the only person who has attended every North American SIGGRAPH conference.

*Moderator - Daniel Cardoso Llach.* Daniel Cardoso Llach is an architect, design scholar, and researcher interested in issues of automation in design, interdisciplinary creativity, human-machine interaction, and technological cultures in architecture and design. He teaches architecture in the School of Architecture at Carnegie Mellon University as an Associate Professor, where he chairs the Master of Science in Computational Design program and co-directs the CoDe Lab. Cardoso Llach's work explores problems ranging from social and cultural aspects of automation in design, the politics of representation and participation in software, and new methods

for using data to visualize design as a socio-technical phenomenon. His recent research includes a book on the cultural history of Computer-Aided Design and numerically controlled machinery

that examines how postwar technological projects shaped conceptions of design informing current architectural practices: *Builders of the Vision: Software and the Imagination of Design* (Routledge, 2015).